

Sanjilraj Arulthasan

Computer Engineering Undergraduate

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ABOUT ME

Proactive Computer Engineering undergraduate with a strong foundation in full-stack development, QA, and DevOps. Proficient in CI/CD with GitHub Actions, Jenkins, and Azure DevOps, with hands-on expertise in MySQL and NoSQL databases. I bring strong problem-solving and teamwork skills to any project. Passionate about creating efficient software solutions and dedicated to continuous learning in the tech industry. Seeking an Industrial Training (Internship) as a **Software Engineer**.

EDUCATION

B.Sc. (Hons) in Engineering (Computer Engineering) May 2023 – Present

University of Ruhuna, Faculty of Engineering

Current Standing: 3rd Year Undergraduate.

Relevant Coursework: Data Structures & Algorithms, Software Engineering, Database Systems, Machine Learning, Operating Systems.

G.C.E. Advanced Level (Physical Science Stream) Aug 2019 – Feb 2022

Vavuniya Tamil Madya Maha Vidyalaya

Results: 2A, 1B

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C, C++, C#, Dart, JavaScript.
- **Web & Mobile:** PERN Stack (PostgreSQL, Express.js, React.js, Node.js), React.js, Spring Boot, Flutter, Firebase, React Native, REST APIs.
- **DevOps & Cloud:** Azure DevOps, Docker, Jenkins, Terraform, Ansible, AWS, CI/CD Pipelines.
- **Databases:** MySQL, MongoDB, PostgreSQL, Neo4j.
- **Testing & QA:** Selenium (UI), JUnit (TDD), Cucumber (BDD), JMeter, SonarQube.
- **Developer Tools:** Git, GitHub, VS Code, IntelliJ IDEA, Android Studio, Jira, Confluence, Azure devops.

PROJECT EXPERIENCE

1. Indoor Stall Pre-Booking System May 2025 – Oct 2025

Tech Stack: React.js, Spring Boot, MySQL

- Architected a full-stack booking platform with a React-based frontend and a Spring Boot REST API backend.
- Implemented secure authentication using JWT and BCrypt alongside Role-Based Access Control (RBAC).
- Developed business logic to prevent time-slot overlaps and manage real-time resource availability.

2. Outcome-Based Quality Assurance System Sep 2025 – Present

Tech Stack: React.js, Spring Boot, MySQL, Chart.js

- Developing an automated platform to calculate and visualize OBE framework attainment levels (LO/PO).
- Implemented complex algorithms for student performance mapping and integrated Chart.js for data visualization.

3. Food Ordering System (Mobile Application) Oct 2025 – Jan 2026

Tech Stack: Flutter, Dart, Firebase

- Developed a cross-platform mobile application with dedicated interfaces for Customers and Administrators.
- Leveraged Firebase Firestore for real-time order synchronization and secure cloud-based authentication.

4. Employee Details Management System (Database Research) June 2025 – July 2025

Tech Stack: MySQL, MongoDB, Neo4j

- Evaluated data modeling strategies across Relational, Document, and Graph database systems.
- Conducted benchmarking to analyze query performance and data retrieval efficiency across different storage engines.

5. GPA Based Ranking System (DSA) Dec 2024 – Feb 2025

Tech Stack: C#, Data Structures

- Built a high-performance ranking engine using a custom-implemented Doubly Linked List for record management.
- Optimized GPA computation and ranking algorithms to achieve $O(n)$ time complexity.

6. Study Hall Management System (Desktop/GUI) Jan 2025 – Apr 2025

Tech Stack: C#, .NET, SQLite

- Engineered a .NET desktop application utilizing the MVVM architecture for scalable and maintainable code.

- Integrated SQLite for local data persistence, ensuring high performance for offline resource management.

7. Speech Emotion Detection (Deep Learning)

Feb 2026 – May 2026

Tech Stack: Python, Keras, TensorFlow, Librosa, LSTM

- Implementing a multi-layered LSTM neural network to classify emotional states from acoustic time-series data.
- Utilized Librosa for feature engineering, extracting MFCCs to capture non-linear human auditory perceptions.

8. DevOps Implementation: Study Hall Infrastructure

Oct 2025 – Feb 2026

Tech Stack: Azure DevOps, Docker, Jenkins, Terraform, Ansible, AWS

- Automated the software development lifecycle by designing robust CI/CD pipelines using Jenkins and Azure DevOps.
- Managed cloud infrastructure using Terraform (IaC) and Ansible for automated server provisioning on AWS.
- Configured Azure Boards and Repos for sprint planning, work-item tracking, and version control governance.

9. Quality Assurance & Testing Framework

Oct 2025 – Jan 2026

Tech Stack: Selenium, Cucumber, JUnit, JMeter

- Developed comprehensive test suites using Selenium and Cucumber for Behavior-Driven Development (BDD).
- Conducted performance and load testing using JMeter to ensure system stability under high-traffic scenarios.

10. Medical Insurance Charges Prediction (ML)

Oct 2025 – Feb 2026

Tech Stack: Python, Scikit-Learn, Random Forest

- Developed a supervised learning model to forecast medical costs based on demographic and health factors.
- Achieved an R^2 score of 0.865 using a Random Forest Regressor after comparative model analysis.

11. Networked Multiplayer Tic-Tac-Toe

Dec 2025 – Feb 2026

Tech Stack: Java, TCP/IP Sockets, Swing, Multithreading

- Built a real-time multiplayer system using Java TCP/IP sockets for synchronized game state management.
- Utilized multithreading to handle concurrent network I/O without interrupting the GUI responsiveness.

12. Full-Stack Task Management Platform (PERN Stack)

Mar 2026 – Present

Tech Stack: PostgreSQL, Express.js, React.js, Node.js

- Built a full-stack task management application using the PERN stack with a RESTful API architecture.
- Designed normalized PostgreSQL schemas and implemented JWT-based authentication and authorization.
- Deployed the application using CI/CD pipelines with containerized Node.js services.

NON-TECHNICAL SKILLS

- **Project Management:** Workload Balancing, Self-motivated, Requirement Analysis.
- **Leadership:** Team Player, Interpersonal Skills, Communication.
- **Professional Skills:** Problem-Solving, Troubleshooting, Adaptability.

ACHIEVEMENTS

- **Global Rank 1075** in IEEEExtreme 19.0 (Top 12% of 9000+ teams).
- **Global Rank 3286** in IEEEExtreme 18.0.

LEADERSHIP & VOLUNTEERING

- **Vice Secretary** – All University Students Development Association, Vavuniya (2024–2025).
- IEEE Student Member | IESL Student Member.
- Organized and coordinated community development events, leading student volunteer teams.
- Volunteered as a tutor for students during the “Mehawara” seminar series.
- **Interests:** Recreational Cricket, Strategic Team Sports.

REFEREES

- **Dr. Rajitha Udawalpola**

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